Empirical Methods in Finance

Methods and Applications

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Introduction

Cont (2001) — Stylized facts on asset returns

Cont, R. (2001), Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues. *Quantitative Finance*

1. **Absence of autocorrelations:** (linear) autocorrelations of asset returns are often insignificant, except for very small intraday time scales ($\sim$ 20 minutes) (microstructure effects).
2. **Heavy tails:** the (unconditional) distribution of returns seems to display a power-law or Pareto-like tail, with a tail index which is finite, higher than two.
3. **Gain/loss asymmetry:** one observes large drawdowns in stock prices and stock index values but not equally large upward movements.
4. **Aggregational Normality:** as one increases the time scale over which returns are calculated, their distribution looks more and more like a normal distribution. The shape of the distribution is not the same at different time scales.
5. **Intermittency:** returns display, at any time scale, a high degree of variability. This is quantified by the presence of irregular bursts in time series of a wide variety of volatility estimators.

Cont (2001) — Stylized facts on asset returns (cont.)

6. **Volatility clustering:** different measures of volatility display a positive autocorrelation over several days, which quantifies the fact that high-volatility events tend to cluster in time.
7. **Conditional heavy tails:** even after correcting returns for volatility clustering (e.g. via GARCH-type models), the residual time series still exhibit heavy tails. However, the tails are less heavy than in the unconditional distribution of returns.
8. **Slow decay of autocorrelation in absolute returns:** the autocorrelation function of absolute returns decays slowly as a function of the time lag, roughly as a power law. This is sometimes interpreted as a sign of long-range dependence.
9. **Leverage effect:** most measures of volatility of an asset are negatively correlated with the returns of that asset.
10. **Volume/volatility correlation:** trading volume is correlated with volatility.
11. **Asymmetry in time scales:** coarse-grained measures of volatility predict fine-scale volatility better than the other way round.

Moments

Moments of a random variable

Definition: The $\boldsymbol{k^{th}}$ non-central moment of $\boldsymbol{X}$

The $\boldsymbol{k}^{th}$ **non-central moment** of X defines as
$m_k = E\left[X^k\right] = \int_{-\infty}^{\infty} x^k f_X(x)\, dx$ for $k = 1, 2, \ldots$

The first non-central moment $m_1 = \mu$ is the expected value of X.

Definition: The $\boldsymbol{k^{th}}$ central moments

The $\boldsymbol{k}^{th}$ **central moments** of X as

$$\mu_k = E\left[(X - m_1)^k\right] = \int_{-\infty}^{\infty} (x - m_1)^k f_X(x)\, dx \qquad (1)$$

for $k = 1, 2, \ldots$

Moments of a random variable

Key points:

- By construction, $\mu_1 = 0$.
- The second central moment $\mu_2 \equiv \sigma^2 = m_2 - \mu^2$ is the variance of X.
- The 3^{rd} central moment $\mu_3 = E\left[(X - m_1)^3\right]$ measures the asymmetry of the distribution
- The 4^{th} central moment $\mu_4 = E\left[(X - m_1)^4\right]$ measures its tail-fatness.

Skewness and kurtosis

Standardized skewness and **kurtosis** are defined as

$$S[X] = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = \frac{\mu_3}{\sigma^3} K[X] = E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] = \frac{\mu_4}{\sigma^4} \tag{2}$$

- When $S[X]$ is negative, large realizations of X are more often negative than positive, suggesting that crashes are more likely to occur than booms.
- A large kurtosis $K[X]$ implies that large realizations (either positive or negative) are more likely to occur.
- For the normal distribution, skewness is equal to zero, while standardized kurtosis is equal to 3. We define the excess kurtosis as $K[X] - 3$.

Measures of location (1/2)

- Now, we consider a time series of log-returns $\{r_t\}$, for $t = 1, \ldots, T$, that we assume to be the realizations of a random variable.
- The **sample mean** (or average) is the simpler estimate of location.

$$\bar{r} = \hat{\mu} = \frac{1}{T} \sum_{t=1}^{T} r_t \tag{3}$$

- If the data sample is drawn from a normal distribution, then the sample average is the optimal measure of location.

Measures of location (2/2)

- The **median** is the 50^{th} percentile of the sample, or 50% of the sample has a value lower than $med = med[r_t]$, i.e.

$$P[r_t \leq \text{med}] = Pr[r_t \geq \text{med}] = \frac{1}{2} \tag{4}$$

- The mean is sensitive to outliers. One bad data value can move the average away from the centre of the rest of the data.
- The median is robust to outliers.

Measures of dispersion (1/2)

- The **variance** is a popular measure of dispersion. It is the optimal measure for normal returns.
- The **standard deviation** is the square root of the variance.

$$s^2 = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \hat{\mu})^2 \qquad \text{or} \qquad \hat{\sigma}^2 = \frac{1}{T}\sum_{t=1}^{T}(r_t - \hat{\mu})^2 \tag{5}$$

- s^2 is the unbiased estimator, while $\hat{\sigma}^2$ is the ML estimator.
- The variance is not robust to outliers.

Measures of dispersion (2/2)

- The **Median of Absolute Deviation from the median** (MAD) is computed as:

$$MAD = \mathrm{med}\left(|r_t - med|\right) \tag{6}$$

- Under normality, the standard deviation is equal to $1.4826 \times MAD$.
- The **Interquartile Range** (IQR) is the difference between the 75^{th} and 25^{th} percentiles.
- The MAD and the IQR are robust to outliers.

Higher moments (1/2)

- The sample counterpart of the standardized skewness and kurtosis are given by

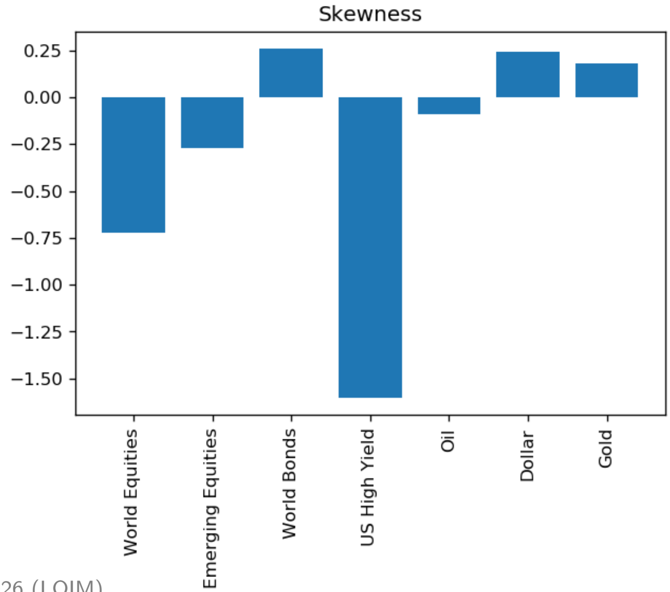
$$\hat{S} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \bar{r}}{\hat{\sigma}}\right)^3, \qquad \hat{K} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \bar{r}}{\hat{\sigma}}\right)^4 \tag{7}$$

- For a normal distribution, the skewness and the excess kurtosis are zero.

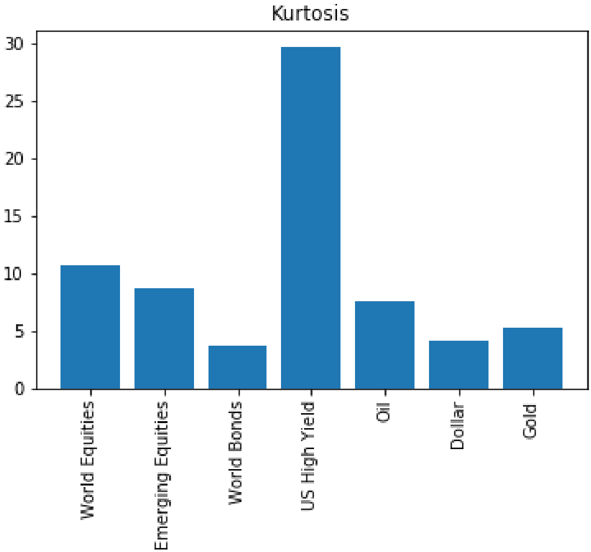
Higher moments (2/2)

- If $\hat{S} < 0$, the distribution has a fatter left tail.
- If $\hat{S} > 0$, the distribution has a fatter right tail.
- If $\hat{K} < 3$, the distribution has thinner tails than the normal.
- If $\hat{K} > 3$, the distribution has fatter tails than the normal.
- **Remark:** If we reject normality, we should probably use robust measures for higher moments.

Asset Returns' Skewness



Asset Returns' Kurtosis



Testing for Normality

Tests for normality

- We consider **unconditional normality** for the moment, i.e., the distribution of the return series itself.
- Several tests for the null hypothesis of normality have been developed, focusing on different implications of the normality assumption:
  - the **moments of the distribution**
    - Jarque-Bera test
  - the **properties of the empirical distribution function**
    - Kolmogorov-Smirnov test
    - Anderson-Darling test

Jarque-Bera Test (1/3)

- The **Jarque-Bera test** is based on the fact that skewness and excess kurtosis are jointly equal to zero under normality.
- Under normality, the first 4 moments have the following asymptotic distributions:

$$\sqrt{T}\,(\hat{\mu}-\mu) \sim N(0,\sigma^2) \qquad \sqrt{T}\,(\hat{\sigma}^2-\sigma^2) \sim N(0,2\sigma^4) \tag{8}$$

$$\sqrt{T}\,(\hat{S}-0) \sim N(0,6) \qquad \sqrt{T}\,(\hat{K}-3) \sim N(0,24) \tag{9}$$

- **Remarks:**
  - Since skewness and kurtosis are already standardized, their distribution does not depend on the mean and/or variance.
  - Skewness and kurtosis will be informative only for a large number of observations.

Jarque-Bera Test (2/3)

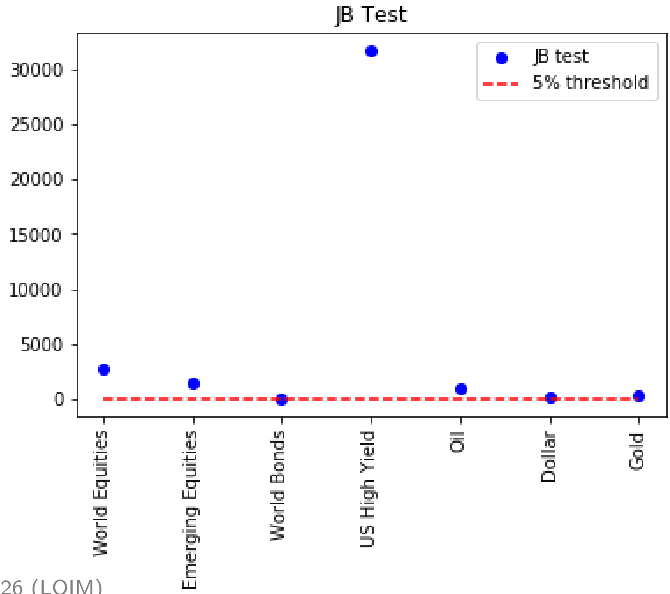
- The null hypothesis of the test is: Skewness $=0$ and Kurtosis $=3$.
- Since the normal distribution is defined by the first two moments, there is no constraint on the mean and variance under the null.
- Under normality, the sample skewness and kurtosis are mutually independent.
- The JB test statistic is simply defined as:

$$JB = T\left[\frac{\hat{S}^2}{6} + \frac{(\hat{K}-3)^2}{24}\right] \tag{10}$$

Jarque-Bera Test (2/3)

- Under the null hypothesis, it is asymptotically distributed as a χ^2.
- If $JB \geq \chi^2_{1-\alpha}(2)$, then the null hypothesis has to be rejected at level α.
- **Example:** If the JB statistic is larger than 6, we should reject the normality hypothesis with only 5% of chances of being wrong.

Jarque and Bera Test



Empirical Distribution Function Goodness-of-Fit Tests (1/2)

- Empirical distribution function goodness-of-fit test compare the empirical cdf and the assumed theoretical cdf $F^*(x;\theta)$ (here, the normal distribution).
- The data consist of a time series $(r_1, r_2, ..., r_T)$ associated with some unknown cdf, denoted by $F_r(.)$.
- As the true distribution $F_r(.)$ is unknown, it is approximated by the empirical cdf, denoted $G_T(.)$, defined as:

$$G_T(x) = \frac{1}{T}\sum_{t=1}^{T} 1_{\{r_t \leq x\}} \tag{11}$$

$G_T(.)$ is a step function that steps of height $1/T$ at each observation.

Empirical Distribution Function Goodness-of-Fit Tests (2/2)

The empirical cdf $G_T(x)$ is compared with the assumed distribution function $F^*(x;\theta)$ to see if there is good fit.

- Null hypothesis $H_0 : F_r(x) = F^*(x;\theta)$ for all x.
- Alternative hypothesis $H_a : F_r(x) \neq F^*(x;\theta)$ for at least one value of x.
- EDF tests are based on the result that if $F_r(X)$ is the cdf of X, then the variable $F_r(X)$ is uniformly distributed between 0 and 1.

Kolmogorov-Smirnov Test (1/2)

- One of the simplest measures of the difference between two cdfs is the largest distance between the two functions $G_T(x)$ and $F^*(x;\theta)$.
- This is the KS test statistic suggested by Kolmogorov (1933), defined as:

$$KS = \max_{t=1,\cdots,T} |G_T(x) - F^*(x;\theta)| \tag{12}$$

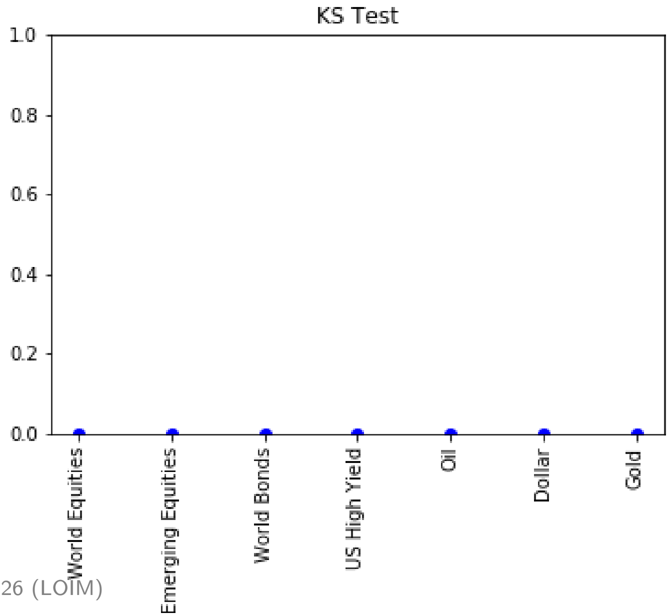
Kolmogorov-Smirnov Test (2/2)

- In practice, this test is very easy to implement:
  - Sort the sample data by increasing order and denote the new sample $\{\tilde{r}_t\}_{t=1}^T$, with $\tilde{r}_1 \leq \cdots \leq \tilde{r}_T$. Then, by construction, $G_T(\tilde{r}_t) = \frac{t}{T}$.
  - Evaluate the assumed theoretical cdf $F^*(\tilde{r}_t; \theta)$ for all values $\{\tilde{r}_t\}_{t=1}^T$. In the case of the normal distribution, we assume that the mean μ and variance σ^2 are known.
  - Compute the KS test statistic:

$$KS = \max_{t=1,\cdots,T} \left| F^*(\tilde{r}_t) - \frac{t}{T} \right| \tag{13}$$

- The critical values of this statistic have been tabulated.

KS Test


KS Test
1.0
0.8
0.6
0.4
0.2
0.0
World Equities
Emerging Equities
World Bonds
US High Yield
Oil
Dollar
Gold

Serial Correlation

- We now consider if the moments of a given series are time dependent. This is an important issue for testing market efficiency.
- Suppose the time series $(r_1, r_2, ..., r_T)$ is **weakly (or covariance) stationary**, so that:

$$E[r_t] = \mu \quad \forall t \tag{14}$$

$$V[r_t] = \sigma^2 \quad \forall t \tag{15}$$

$$Cov[r_t, r_{t-k}] = \gamma_k \quad \forall t,\ k \neq 0 \tag{16}$$

- The **auto-covariance** γ_k measures the dependency of r_t with respect to its own past observation r_{t-k}.

Serial correlation (2/3)

- We have the following relations:

$$\gamma_0 = V[r_t] = E[(r_t - \mu)^2] \tag{17}$$

$$\gamma_k = \gamma_{-k} \tag{18}$$

- For a **weakly stationary** process, the autocovariance γ_k depends on the horizon k, but not on the date t.
- Since the auto-covariance is an unbounded measure of dependency, we generally prefer the **auto-correlation function**.

$$\rho_k = \frac{Cov[r_t, r_{t-k}]}{\sqrt{V[r_t]V[r_{t-k}]}} = \frac{Cov[r_t, r_{t-k}]}{V[r_t]} = \frac{\gamma_k}{\gamma_0} \tag{19}$$

- with $\rho_k = \rho_{-k}$, $\rho_0 = 1$ and $-1 \leq \rho_k \leq 1$.
- A consistent estimator of ρ_k is:

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{T}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{T}(r_t - \bar{r})^2} \quad \text{for} \quad 0 < k < T - 1 \tag{20}$$

- Since $\hat{\rho}_k$ is a ratio of quadratic functions of r_t, its sample properties are not easily obtained.
- We have to consider large sample properties.
- We have a simple expression for the asymptotic distribution in some special cases.

Serial correlation (asymptotic results)

- $\rightarrow$ If r_t is a **i.i.d.** process, then $V[\hat{\rho}_k] = 1/T$, so that

$$\sqrt{T}\,\hat{\rho}_k \text{ is asymptotically normal } N(0,1),\ \forall k > 0 \tag{21}$$

- $\rightarrow$ If r_t is a **stationary moving average** $MA(q)$ process with

$$r_t = \mu + \sum_{i=0}^{q} \theta_i \varepsilon_{t-i} \tag{22}$$

- with $\theta_0 = 1$ and $\rho_k = 0$ for $k > q$, then

$$V[\hat{\rho}_k] = \frac{1}{T}\left(1 + 2\sum_{i=1}^{q} \rho_i^2\right) \tag{23}$$

- so that $\sqrt{T}\,\hat{\rho}_k$ is asymptotically normal $N(0, V)$ with $V = 1 + 2\sum_{i=1}^{q} \rho_i^2$ for all $k > q$.
- **Idea:** For the test $\rho_k = 0$, we assume that the process is an $MA(k-1)$ to estimate $V[\hat{\rho}_k]$.

Box-Pierce / Ljung-Box tests (1/2)

- We now want to test the null hypothesis that the p first serial correlations are equal to 0,

$$H_0 : \rho_1 = \cdots = \rho_p = 0 \text{ vs } H_a : \rho_k \neq 0 \text{ for some } k = 1, \ldots, p. \tag{24}$$

- A simple test can be based on the **Box-Pierce Q-statistic**:

$$Q_p = T \sum_{j=1}^{p} \hat{\rho}_j^2 \tag{25}$$

Box-Pierce / Ljung-Box tests (2/2)

- Under $H_0 : \rho_1 = \cdots = \rho_p = 0$, all the $\hat{\rho}_k$ have $V[\hat{\rho}_k] = 1/T$.
- Therefore, the Q_p statistic is asymptotically distributed as $\chi^2(p)$.
- If $Q_p \geq \chi^2_{1-\alpha}(p)$, then the null hypothesis has to be rejected at level α.
- A common practice consists in testing the nullity of the p first serial correlations for different values of p.
- The **Ljung-Box Q'-statistic**:

$$Q'_p = T(T+2)\sum_{j=1}^{p}\frac{1}{T-j}\hat{\rho}_j^2 \tag{26}$$

- has better finite-sample properties, with the same asymptotic distribution.

Serial correlation of squared returns

Index	World Equities	Emerging Equities	World Bonds	US High Yield	Oil	Dollar	Gold	Critical Value
0	0	0	0	0	0	0	0	3.84146
1	108.086	99.5734	43.2306	139.803	135.913	39.2922	18.1…	5.99146
2	184.836	161.994	62.6456	155.819	391.365	61.9924	33.0…	7.81473
3	261.384	372.555	65.4849	168.146	547.704	84.3125	40.5…	9.48773
4	294.538	433.862	66.5476	207	667.146	85.946	81.9…	11.0705
5	314.105	467.85	74.781	228.775	825.415	88.7527	152.…	12.5916
6	342.667	505.184	76.709	278.875	876.715	92.5027	161.…	14.0671
7	400.215	563.331	78.3262	295.924	958.483	106.8	168.…	15.5073
8	410.331	581.011	88.7806	298.822	1012.42	121.863	174.…	16.919

Serial correlation in volatility (1/2)

- To test for time dependency in volatility, we need a time-varying measure of volatility.
- There are at least two possible measures:
  - (1) the mean-adjusted squared returns
  - (2) the absolute returns
- Assume that returns have the following dynamics:

$$r_t = \mu + \varepsilon_t \qquad \text{with} \qquad \varepsilon_t = \sigma_t z_t \tag{27}$$

- where μ is the constant mean, ε_t the unexpected returns, σ_t the time-varying volatility based on information at date $t-1$ and z_t is an $N(0,1)$ innovation.
- **Remark:** We will have a session devoted on the modeling of the conditional volatility.

Serial correlation in volatility (2/2)

- Then with information set I_{t-1} at time $t-1$, we have:

$$E\left[\varepsilon_t^2 \mid I_{t-1}\right] = E\left[\sigma_t^2 z_t^2 \mid I_{t-1}\right] = \sigma_t^2 E\left[z_t^2 \mid I_{t-1}\right] = \sigma_t^2 \quad (28)$$

- because z_t^2 is distributed as a $\chi^2(1)$.
- Therefore, ε_t^2 can be viewed as a proxy for the volatility at time t.
- Alternatively, we have $\varepsilon_t \sim N(0, \sigma_t^2)$. Then:

$$E[|\varepsilon_t| \mid I_{t-1}] = \sigma_t \sqrt{2/\pi} \quad (29)$$

- Consequently, $|\varepsilon_t| / \sqrt{2/\pi}$ can be used as a proxy for σ_t.
- These two measures are noisy estimators of conditional volatility.

Serial correlation of squared returns

Index	World Equities	Emerging Equities	World Bonds	US High Yield	Oil	Dollar	Gold	Critical Value
0	0	0	0	0	0	0	0	3.84146
1	108.086	99.5734	43.2306	139.803	135.913	39.2922	18.1...	5.99146
2	184.836	161.994	62.6456	155.819	391.365	61.9924	33.0...	7.81473
3	261.384	372.555	65.4849	168.146	547.704	84.3125	40.5...	9.48773
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5	314.105	467.85	74.781	228.775	825.415	88.7527	152....	12.5916
6	342.667	505.184	76.709	278.875	876.715	92.5027	161....	14.0671
7	400.215	563.331	78.3262	295.924	958.483	106.8	168....	15.5073
8	410.331	581.011	88.7806	298.822	1012.42	121.863	174....	16.919

Cross-asset correlation

Cross-correlation (1/3)

- A natural way to measure the dependence between two series is the correlation.
- We first define the **covariance** between two series $r_{1,t}$ and $r_{2,t}$:

$$Cov[r_{1,t}, r_{2,t}] = \frac{1}{T}\sum_{t=1}^{T}(r_{1,t} - \mu_1)(r_{2,t} - \mu_2) \tag{30}$$

- where μ_1 and μ_2 denote the sample means of $r_{1,t}$ and $r_{2,t}$.
- The interpretation of the covariance is not easy, because it depends on the level of variances.
- Then, we define the **correlation** between $r_{1,t}$ and $r_{2,t}$:

$$\rho[r_{1,t}, r_{2,t}] = \frac{Cov[r_{1,t}, r_{2,t}]}{\sqrt{V[r_{1,t}]V[r_{2,t}]}} \tag{31}$$

Cross-correlation (2/3)

- By construction, $\rho[r_{1,t}, r_{1,t}] = 1$, $\rho[r_{1,t}, r_{2,t}] = \rho[r_{2,t}, r_{1,t}]$ and $-1 \leq \rho[r_{1,t}, r_{2,t}] \leq 1$.
- It is worth noting that the correlation is only a measure of **linear association**.
- Consequently, we may find that two series are uncorrelated, while they are not independent.
- The correlation between $r_{1,t}$ and $r_{2,t}$ is estimated as:

$$\hat{\rho}[r_{1,t}, r_{2,t}] = \frac{\sum_{t=1}^{T}(r_{1,t} - \hat{\mu}_1)(r_{2,t} - \hat{\mu}_2)}{\sqrt{\sum_{t=1}^{T}(r_{1,t} - \hat{\mu}_1)^2 \ \sum_{t=1}^{T}(r_{2,t} - \hat{\mu}_2)^2}} \quad (32)$$

Cross-correlation (3/3)

Index	World Equities	erging Equit	World Bonds	JS High Yiel‹	Oil	Dollar	Gold
World Equities	1	0.803145	-0.00132748	0.589241	0.293068	-0.330414	0.115807
Emerging Equities	0.803145	1	0.0415657	0.562176	0.314477	-0.353431	0.223592
World Bonds	-0.00132748	0.0415657	1	0.0604445	0.0171112	-0.719327	0.441196
US High Yield	0.589241	0.562176	0.0604445	1	0.263098	-0.237519	0.0840394
Oil	0.293068	0.314477	0.0171112	0.263098	1	-0.198737	0.187811
Dollar	-0.330414	-0.353431	-0.719327	-0.237519	-0.198737	1	-0.483749
Gold	0.115807	0.223592	0.441196	0.0840394	0.187811	-0.483749	1

Characteristics of Financial Asset Returns

- However, for actual returns, these assumptions have been rejected in several ways:
  - **Normality**
    - Asymmetry
    - Fat tails
    - Aggregated normality
  - **Time independency**
    - No serial correlation
    - Volatility clustering
    - Asymmetric volatility
  - **Multivariate dependency**
    - Time-varying correlation

Volatility asymmetry (1/2)

- Another important feature of financial market volatility is that it is more affected by negative returns than by positive returns.
- To illustrate the point, we perform the following regressions:

$$\varepsilon_t^2 = \omega + \beta\varepsilon_{t-1}^2 + \gamma\varepsilon_{t-1}^2 \times 1_{\{\varepsilon_{t-1}<0\}} \tag{33}$$

or

$$|\varepsilon_t| = \omega + \beta\,|\varepsilon_{t-1}| + \gamma\,|\varepsilon_{t-1}| \times 1_{\{\varepsilon_{t-1}<0\}} \tag{34}$$

- where β measures the direct effect of past returns, and γ captures the additional impact of negative return shocks.

Volatility asymmetry (2/2)

	S&P500	Nikkei	DAX	FTSE100
Squared returns				
ω	17.633	26.912	32.032	18.354
(std err.)	2.237	3.190	3.997	2.782
β	-0.122	0.036	-0.044	0.057
(std err.)	0.117	0.087	0.102	0.142
γ	0.299	0.246	0.217	0.043
(std err.)	0.121	0.094	0.108	0.145
Absolute returns				
ω	2.854	3.781	3.845	2.912
(std err.)	0.199	0.264	0.274	0.211
β	-0.022	0.043	0.014	0.065
(std err.)	0.064	0.062	0.064	0.066
γ	0.255	0.163	0.203	0.113
(std err.)	0.064	0.062	0.064	0.065